

FINAL YEAR PROJECT PROPOSAL

Project I (3.0 Credits) & Project II (3.0 Credits)

Design and Implementation of a Secure Three-Tier Banking Wide Area Network (BankWAN) for Xeonipi Bank Ltd. Using Cisco Enterprise Networking and Windows Server 2022

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01. Executive Summary

Modern banking organizations require secure, reliable, scalable, and centrally managed communication infrastructure to connect their Head Office, regional Zone Offices, and Branch Offices. This project proposes the design and implementation of a **three-tier Banking Wide Area Network (BankWAN) for Xeonipi Bank Ltd.**, consisting of Head Office (Dhaka), Zone Offices, and Branch Offices.

The proposed network will first be designed, configured, and validated using **Cisco Packet Tracer** as a complete network simulation environment. The simulation will demonstrate the complete **Head Office → Zone Office → Branch Office** communication architecture, including VLAN segmentation, Inter-VLAN Routing, routing, Access Control Lists (ACLs), HSRP-based redundancy, and WAN connectivity.

Following the network simulation, a separate virtualized laboratory environment will be implemented using **Windows Server 2022**. Windows Server 2022 will act as the centralized enterprise service platform, providing Active Directory Domain Services (AD DS), DNS, DHCP, centralized user and group management, Organizational Units, Group Policy, and domain-joined Windows clients.

The Cisco Packet Tracer and Windows Server 2022 environments will remain **separate environments**. Packet Tracer will demonstrate and validate the BankWAN network architecture, while Windows Server 2022 will demonstrate the practical enterprise service and centralized identity-management layer based on the proposed architecture.

2. Background and Problem Statement

A banking organization requires continuous and secure communication between its central office, regional offices, and branches. A flat or poorly segmented network can expose critical systems to unauthorized access, excessive lateral movement, uncontrolled inter-department communication, and operational disruption.

To address these challenges, Xeonipi Bank Ltd. will be designed using a hierarchical three-tier architecture:

Head Office → Zone Office → Branch Office

The Head Office will operate as the central control and enterprise service location. Zone Offices will aggregate regional branch connectivity, while Branch Offices will provide customer-facing banking operations.

Network segmentation will be achieved through dedicated VLANs for Teller/Cash, Accounts/Loan, HR, IT, Guest, Voice, and CCTV services. ACLs will control communication between network segments, while HSRP will provide gateway redundancy at the Head Office.

Windows Server 2022 will provide centralized enterprise services and identity management, allowing users and computers to be managed through a central domain environment.

3. Project Objectives

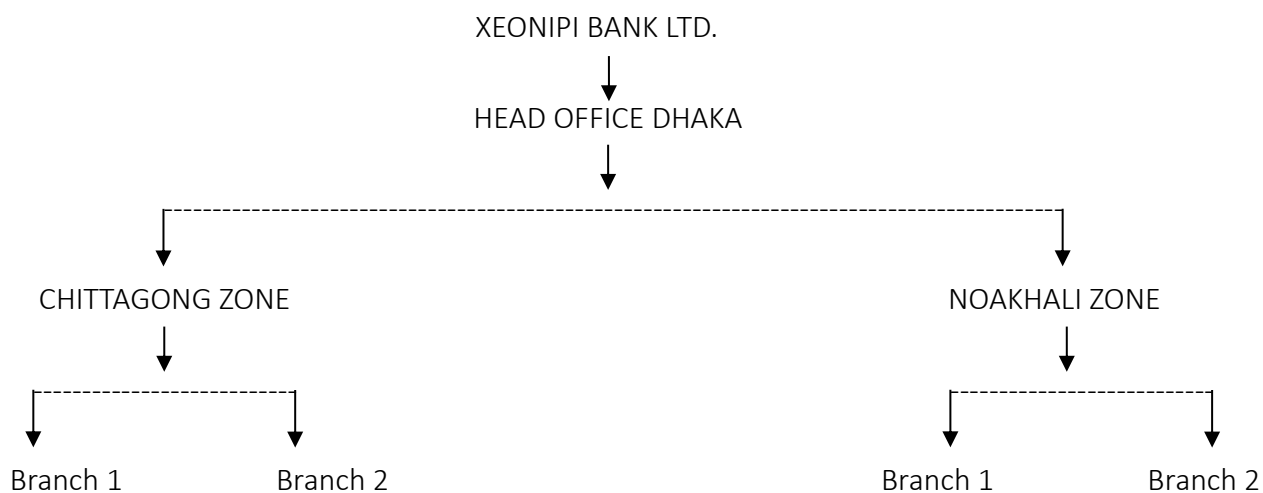
The primary objective is to design and demonstrate a secure, scalable, and centrally managed banking network infrastructure.

The specific objectives are to:

- Design a three-tier BankWAN architecture for Head Office, Zone Offices, and Branch Offices.
- Implement VLAN-based network segmentation.
- Configure Inter-VLAN Routing and enterprise routing.
- Establish secure logical communication between Head Office, Zones, and Branches.
- Implement ACL-based network access control.
- Implement HSRP for gateway redundancy at Head Office.
- Develop a scalable network model for adding future branches.
- Deploy Windows Server 2022 as the centralized enterprise service platform.
- Implement Active Directory Domain Services, DNS, and DHCP.
- Create centralized users, groups, and Organizational Units.
- Join Windows clients to the central domain.
- Implement selected Group Policy security controls.
- Validate the proposed infrastructure through practical simulation and laboratory demonstrations.

4. Proposed System Architecture

The BankWAN will follow a hierarchical three-tier structure.



Head Office – Dhaka

The Head Office will serve as the central network and enterprise service location. It will contain redundant Layer-3 Core Switches, WAN connectivity, centralized server infrastructure, and enterprise network segments.

Major VLANs will include:

- VLAN 10 – Teller/Cash
- VLAN 20 – Accounts/Loan
- VLAN 30 – HR
- VLAN 40 – IT
- VLAN 50 – Guest
- VLAN 60 – Voice
- VLAN 100 – CCTV
- Server and Management segments

HSRP will be configured between the Core Layer-3 switches to provide gateway redundancy.

Zone Offices

Zone Offices, such as Chittagong and Noakhali Zones, will operate as regional aggregation points. Each Zone will contain a Layer-3 switch, WAN router, wireless infrastructure, and regional network services where required.

Branch Offices

Branch Offices, such as Agrabad and Maijdee Branches, will contain a WAN router, Layer-2 switch, wireless access point, client devices, IP phones, and IP cameras. Branches will not require dedicated central enterprise servers; instead, they will access centralized services through the proposed WAN architecture.

5. Implementation Methodology

The project will be implemented through **two separate but complementary environments**.

Part A – Cisco Packet Tracer: BankWAN Simulation

Cisco Packet Tracer will be used to design, configure, simulate, and validate the complete BankWAN network.

The demonstration will cover:

- Head Office network.
- Zone Office networks.
- Branch Office networks.
- VLAN configuration and trunking.
- Inter-VLAN Routing.
- Routing between HO, Zones, and Branches.
- ACL-based network security.
- HSRP-based redundancy.
- WAN connectivity.
- Network troubleshooting and verification.

The primary demonstration flow will be:

Head Office → Zone Office → Branch Office

Connectivity and security will be validated through appropriate Cisco verification commands and network testing.

The final .pkt topology and configuration will be included as a major project deliverable.

Part B – Windows Server 2022: Enterprise Service Laboratory

After validating the network architecture through Cisco Packet Tracer, a separate virtualized environment will be implemented using VirtualBox or VMware.

Windows Server 2022 will operate as the **centralized enterprise service platform**.

The practical lab will demonstrate:

- Active Directory Domain Services (AD DS).
- Domain Controller deployment.
- DNS configuration.
- DHCP configuration.
- Organizational Units.
- User and security group management.
- Domain-joined Windows clients.
- Centralized authentication.
- Group Policy configuration.
- Basic file service and permission management.

An example enterprise domain may be configured as:

xeonipibank.local/xeonipibank.com

The Windows clients will obtain network configuration through DHCP, resolve internal resources through DNS, join the central domain, and authenticate using centrally managed user accounts.

6. Project I and Project II Plan

Project I – 3.0 Credits

BankWAN Network Design, Simulation and Validation

Project I will focus primarily on the Cisco Packet Tracer implementation.

Major activities:

- Requirement analysis.
- Three-tier network architecture.
- IP addressing and subnetting.
- VLAN and trunk configuration.
- Inter-VLAN Routing.

- WAN routing.
- ACL implementation.
- HSRP implementation.
- HO → Zone → Branch connectivity.
- Network testing and troubleshooting.

Expected Deliverables:

- Complete Cisco Packet Tracer .pkt file.
- Network topology.
- IP addressing and VLAN plan.
- Routing and ACL configurations.
- HSRP configuration.
- Simulation and testing documentation.

Project II – 3.0 Credits**Windows Server 2022 Enterprise Service Implementation**

Project II will focus on the practical enterprise service laboratory.

Major activities:

- Windows Server 2022 deployment.
- Active Directory implementation.
- DNS and DHCP configuration.
- OU, user, and group management.
- Windows client domain joining.
- Centralized authentication.
- Group Policy implementation.
- File service and access control demonstration.
- Practical testing and documentation.

Expected Deliverables:

- Windows Server 2022 laboratory environment.
- AD DS, DNS, and DHCP implementation.
- Domain users, groups, and OUs.
- Domain-joined Windows clients.
- GPO configuration.
- Final technical documentation and demonstration.

7. Hardware, Software and Expected Outcome

Recommended Hardware

The practical laboratory will require a computer with approximately Intel Core i7/AMD Ryzen 7 processor, 32 GB RAM, 1 TB SSD storage, and hardware virtualization support (Intel VT-x/AMD-V).

Software

- Cisco Packet Tracer 8.x or available stable version.
- Windows Server 2022.
- Windows 10/11 Enterprise.
- Oracle VirtualBox or VMware.
- Microsoft Word and diagramming software for documentation.

Expected Outcome

The completed project will demonstrate both the **network infrastructure layer** and the **centralized enterprise service layer** of a simulated banking organization.

Cisco Packet Tracer
↓
HO → Zone → Branch
↓
VLAN + Routing + ACL + HSRP
↓
BankWAN Network Simulation

Windows Server 2022
↓
AD DS + DNS + DHCP
↓
Users + Groups + OU + GPO
↓
Windows Clients
↓
Centralized Enterprise Services

8. Conclusion

The proposed **Xeonipi Bank Ltd. BankWAN** project will provide a practical model of a secure and scalable banking infrastructure by combining enterprise networking with centralized server management.

Cisco Packet Tracer will be used to demonstrate and validate the complete **Head Office → Zone Office → Branch Office** network architecture, including VLAN segmentation, routing, ACL-based access control, HSRP, and WAN connectivity. A separate Windows Server 2022 virtual laboratory will then demonstrate centralized enterprise services including Active Directory, DNS, DHCP, user management, Group Policy, and domain-joined clients.